

Dry Eye Relief

Natural Medicine for Accelerated Self-Healing: Starter Excerpt

by Dr. Glen Swartwout, A.B., B.D., O.D., N.D., F.I.C.A.N., F.C.S.O.

What follows is drawn from the front matter and opening chapter of *Dry Eye Relief*, offered here largely as Dr. Glen wrote it, lightly arranged so you can hear directly from him before deciding whether to go further.

Preface

This book began as research notes for a presentation to the International College of Applied Nutrition in Tokyo, Japan, while I was establishing the Optometric Center of Tokyo to bring my father's pioneering work in natural eye and vision care to the Orient.

The book took further form through many years of clinical practice of Holistic Optometry and European Biological Medicine, as well as while co-authoring *Natural Eye Care: An Encyclopedia*, with Dr. Marc Grossman, O.D., L.Ac.

Introduction

The purpose of this book is to provide an overview of insights and resources for potential therapies in the healing of the underlying causes of dry eye syndrome in all its many variations, which are ultimately unique to each person, and to some degree changing in time through the course of the condition and its healing. This is especially true in accelerated self-healing, where causal stress patterns are identified individually in real time, and the body's healing

processes are supported with natural remedies like those introduced in this book, based on functional response testing, which is ultimately a form of biocommunication.

Remedies found to be effective in an individual case will never be limited to those identified in this book, since the causal chains present in the whole person must be treated, not just a presenting or salient symptom, which may well be at the effect or "tail end" of causality.

Nevertheless, for both the clinician and the patient seeking palliative relief of symptoms or remedy possibilities particular to an individual or challenging case, this book is intended to offer an array of options to test and consider. In particular, the natural remedies in this book can readily be applied with adherence to the primary dictum of healing, "Primum non nocere."

Dedication

This book is dedicated to my Apprentice, Rae Luscombe, who knows well the challenges of Dry Eye ever since suffering from Bell's Palsy.

Solutions

Research shows that unpreserved eye drops enhance corneal healing and improve dry eye problems in comparison to preserved products. Fortunately several unpreserved tear substitutes are available, including VIVA drops (which contains 19 I.U. of vitamin A per drop) and several single dose tubes. Concern has been expressed however about the risk of corneal abrasion by the sharp ends of single dose tubes. Among the other problems with eye drops is that 80% of people have difficulty getting them in the eyes, and even with practice 49% continue to have poor aim. Less than a third have even been shown how to do it right! The best

method is to pull down the lower lid so the drop can be instilled onto the lower cul de sac (blind pouch) of the conjunctiva rather than directly onto the cornea. For those who have difficulty, eye drop guides are available to assist. For those who are unable to open the eyes while drops are instilled, drop them into the inner corner of the closed lids, and the drops will enter the eyes on opening. Keep drops refrigerated if you experience stinging on instillation. Be sure to keep your eye drops separate from any similar looking bottles, and to prevent contamination, be very careful not to touch the dropper with anything including fingers or the eye itself. Look for a ring of dirt between the cap and lid as a sign that contamination has probably occurred. Be sure to wash your hands before using eye drops and keep the bottle in a clean place.

Nearly one out of three bottles become contaminated, and should be replaced with a clean, fresh supply.

Blinking and tearing following use of an eye drop can dilute the remedy to less than 1% of its original concentration in just a few minutes. To optimize therapeutic effectiveness after instillation, don't blink, but rather close the eyes for up to 2 minutes. You can even pinch gently at the root of the nose to prevent drainage through the tear ducts, increasing effectiveness by 65%. This is also a perfect time to help the eyes relax and improve circulation to the eyes with techniques like palming, the eye stretch, and acupressure or shiatsu eye massage. Between applications, it is important to blink frequently and fully, as this spreads the tears over the front of the eye, moistening any dry areas. The normal tear break-up time (BUT) is over 10 seconds, permitting that length of staring before the cornea begins to show dry spots. With dry eye syndrome, this often shortens to just a few seconds. Staring also contributes to visual stress and is related to increases in myopia.

Related eye muscle tension may further reduce circulation in the eye area. There are three common hypotheses for the cause of dry eye:

- Dry spots on the ocular surface
- Inflammation of the eye surface
- Elevated tear film osmolarity (salinity) with loss of water from the tear film

Dry spots are associated with but don't cause dry eye, as half of dry eye patients have enough tear film stability to prevent dry spots between blinks. Dry eye surface disease is most severe within the palpebral aperture, but there are also changes on the superior and inferior bulbar conjunctiva, where dry spots do not occur. Rabbit studies show that loss of the corneal cell surface proteins that make the cornea wettable is a late change in dry eye disease.

In the 1800s, Hass theorized that dry eye was caused by a direct immune or inflammatory assault on the eye surface, and that the changes on the eye surface are unrelated to the decrease in tear production. This hypothesis was intriguing until researchers started looking at ocular surface pathology in dry eye, in patients who develop dry eye after having their lacrimal glands removed for epiphora, and in patients with Sjögren's syndrome. The group with iatrogenic disease had exactly the same ocular surface changes as the Sjögren's group.

In rabbits, every ocular surface change in Sjögren's syndrome is manifest in normal rabbits by raising tear film osmolarity, whether by blocking tear secretion, decreasing corneal sensation or inducing meibomian gland dysfunction. Research shows expression of inflammatory mediators (cytokines) on the eye surface in dry eye, but inflammation is a non-specific response to any tissue damage. Inflammation helps clean up the damage. Increases in tear film osmolarity cause the increase of pro-inflammatory cytokines on the eye surface. There is no primary immune-inflammatory attack on the eye surface in dry eye.

When Sjögren's first described the ocular surface pathology in dry eye, he concluded that it looked as if some "force" was pulling

water across the ocular surface epithelium. It turns out that force was the osmotic force created by the osmotic gradient created by an elevation of tear film osmolarity. The ocular surface disease in dry eye is dependent on, and proportional to, elevations in tear film osmolarity. Farris finds that tear film osmolarity in dry eye patients increases through the course of each day.

The Tear Film

There are three layers to the normal tear film that keep the front surface of the eye comfortably lubricated and optically clear. The first is a lipid (oil) layer which prevents tears from evaporating. This is secreted by the meibomian glands. The middle, and thickest layer, is the aqueous or watery layer, secreted by the lachrymal gland. The third layer, which binds the tears to the surface of the eye by making the eye tissue wettable, is a protein layer called mucin. This is secreted directly from the conjunctival surface on the white of the eye by goblet cells.

Dry eye syndrome is often related to other health conditions in the body. It is commonly associated with dryness of other mucus membranes and even interior body surfaces like joints (arthritis). Many dry eye patients also complain of connective tissue problems like brittle nails and many experience skin sensitivity to detergents. Digestive imbalances and autoimmune processes like Sjögren's syndrome and lupus are frequently discovered underlying dry eye symptoms. When dry eye symptoms improve, there is also frequently noted an improvement in sinus and post-nasal drip conditions.

Sjögren's syndrome is considered the most commonly misdiagnosed of all health conditions in women over the age of 40, affecting 4 million Americans. In Sjögren's syndrome, dry eye symptoms are accompanied by dryness in other parts of the body. This cluster of symptoms is produced by an autoimmune process in which antibodies attack fluid-secreting cells. These glandular cells can ulti-

mately be destroyed, and occasionally even become cancerous. Dry eye may be accompanied by fatigue, dry mouth (perhaps with a sore throat and swelling of the parotid gland), dry joints (arthritis), dry skin, dry cracked lips, dry scalp (dandruff), and dry, brittle nails. Lack of saliva can lead to dental caries. Doctors may prescribe anti-inflammatory and immunosuppressive steroids. In advanced cases, cyclosporin may be prescribed as well. Allergy to drugs, especially penicillin, may be linked to Sjögren's syndrome.

Limits of Conventional Care

Dry eye syndrome is one of the most common eye conditions. In 1990, about 33 million Americans experienced varying degrees of dry eye symptoms. By 1997 that figure increased to an alarming 59 million. The symptoms may include dryness, grittiness, irritation, burning and even excessive tearing or watering. While many people find temporary relief with artificial tear preparations, this only palliates the symptoms, and preservatives found in many of these products can even aggravate the condition itself. In fact, these preservatives can kill corneal cells. Recovery of corneal health after exposure to preservatives takes a full week, while the conjunctiva takes up to 2 months to heal. The common eye drop preservative benzalkonium chloride causes worsening of dry eyes, making one more and more dependent on the use of such drops. This preservative, along with thimerosal and Tween, occur in toxic concentrations in eye drops. Other pharmaceutical ingredients in eye drops contribute to dry eyes, too. Vasoconstrictors that "get the red out" reduce circulation in the eye, decrease production of the tear film, and worsen dry eye symptoms.

Many doctors recommend treatments that can include placing silicone plugs to block the eyes' tear ducts and even cauterizing the ducts closed to retain moisture in the eye. While such treatments do reduce the amount of artificial tears needed for many with severely dry eyes, it may merely mean applying drops every 4 hours

or more rather than every 3 hours or less. Humidifiers may be helpful in dry or air conditioned environments, and special moisture chamber goggles can also be worn to minimize tear evaporation. Even neutral gray wrap-around sun goggles may offer some relief especially from aggravation by sun, wind and dust. Side shields reduce evaporation by up to 40%. Perhaps if we look a little deeper at the pathophysiology and biochemistry of the tears and tear production, we will find healthier solutions for many dry eye sufferers.

There is much more waiting for you. This starter excerpt is only the preface, introduction, dedication, and opening chapter of *Dry Eye Relief*. The full book goes on to walk through the rest of Dr. Glen's clinical research, chapter by chapter: risk factors, lifestyle and biological terrain, diet and digestion, neuroprotection, antioxidants, minerals, vitamins, essential fatty acids, botanical allies, homeopathy, and the acupuncture and energetic methods he uses to match each remedy to your own eyes.

If what you've just read speaks to where you are right now, whether you're newly dealing with dry, gritty, burning eyes or you've been fighting this for years, the full book is the next step. And if you'd rather talk it through with a real person before you dive in, Dr. Glen still makes time to sit with people navigating exactly this. Reach out and start that conversation.